

LLM-Guided Relational Transition Graphs for Temporal Knowledge Graph Reasoning

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A Additional Experimental Results

This appendix provides details of the benchmarks used, implementation details, and additional experimental results that were omitted from the main text.

A.1 Dataset Details

The proposed methodology is evaluated on three widely used public TKG benchmarks, representing datasets of varying sizes. The ICEWS14 and ICEWS18 benchmarks comprise subgraphs derived from the Integrated Crisis Early Warning System (ICEWS) [1], while the GDELT dataset comprises a subgraph extracted from the Global Database of Events, Language, and Tone (GDELT) [5]. Both databases record global political events together with their corresponding timestamps and consist of separate chronologically split sets of training, validation and testing sets with the proportions of 80%, 10%, 10%. Table 1 summarises the statistics of the benchmarks, which are categorised as small, medium, or large according to the number of training quadruplets.

Table 1: Statistics of the TKG datasets

Dataset	ICEWS14	ICEWS18	GDELT
Entities	6,869	10,094	7,691
Relations	230	256	240
Timestamps	365	365	2975
Training	74,845	373,018	1,734,399
Validation	8,514	45,995	238,765
Test	7,371	49,545	305,241
Category	Small	Medium	Large

A.2 LLM Prompting and details

The transition graph encodes, for each relation, a top-k set of successor relations (those likely to follow) and inhibitor relations (those suppressed by it), each with an associated confidence weight in $[0, 1]$. The transition graph is constructed offline, once per dataset, via a single LLM query per relation. LLM transition relation mining operates via the

Groq API engine, using the llama-3.1-8b-instant checkpoint. No LLM inference occurs during training or evaluation of the temporal knowledge graph reasoning model. Below is the prompt template.

```
Domain: {domain_hint}
Anchor relation: "{anchor_name}"

Observed examples of "{anchor_name}" in the data:
e.g. {subj_1} -> [{anchor_name}] -> {obj_1}
e.g. {subj_2} -> [{anchor_name}] -> {obj_2}
e.g. {subj_3} -> [{anchor_name}] -> {obj_3}

From the relation list below, select exactly {top_k}
SUCCESSORS and exactly {top_k} INHIBITORS for the
anchor relation.

SUCCESSOR: a relation likely to occur SHORTLY AFTER the
anchor.
Score 1-10: 10 = almost certain to follow, 1 = unlikely
to follow.

INHIBITOR: a relation whose occurrence is SUPPRESSED by
the anchor.
Score 1-10: 10 = strongly suppressed, 1 = weakly
suppressed.

Rules:
- Use ONLY names from the list below, copied EXACTLY.
- Each relation is ONE object: {"relation":"name","score":"N"}
- Never combine two relation names in a single object.

Relation list:
{relation_list}

Return ONLY this JSON, no explanation:
{"successors":[{"relation":"exact name","score":8},...],
 "inhibitors":[{"relation":"exact name","score":9},...]}
```

Here, $\{domain_hint\}$ denotes a short description of the dataset's event domain, while $\{anchor_name\}$ denotes the name of the target relation. The LLM is provided with up to three training triples involving the anchor relation, thereby constituting a few-shot, grounded prompting approach. $\{relation_list\}$ enumerates all other relations in the dataset, constraining selection to this closed candidate set. top-k retains the k highest-confidence successor and k highest-confidence inhibitor relations per anchor relation. Failed responses are retired until valid JSON containing the top k successor and inhibitor lists is successfully returned.

A.3 Experimental Settings

The base model follows the LogCL [2] framework. Following literature in temporal knowledge graph (TKG) reasoning, the embedding dimension is set to 200 across all datasets, with a learning rate of 0.001. The Adam optimiser is employed to optimise the model parameters. In the LogCL architecture, the R-GCN comprises two layers, each with a dropout rate of 0.2. For all datasets, the number of kernels is set to 50, with each kernel having a size of 2×3 . The prediction weight is set to 0.9, and a dropout rate of 0.2 is applied in the decoder. The temperature coefficient is set to 0.03 for ICEWS14 and ICEWS18 and to 0.07 for GDELT. In addition, the static KG constraints proposed in [6] are applied to the ICEWS datasets. The local history sequence length m is set to 7 for all datasets. The transition $top - k$ is set to 10 for all benchmarks. The hyperparameters λ_{trans} in mechanism A and α in mechanism B are ranged from 0.1 to 0.5. Raw LLM confidence scores (1–10) are linearly rescaled to [0, 1]. All experiments are conducted on a single NVIDIA RTX A5000. This study adopts the same stopping criteria as LogCL, where training stops early if the exponential moving average of validation filtered MRR fails to improve for 5 consecutive epochs.

We evaluate the effectiveness of the proposed approach using the widely adopted temporal knowledge graph reasoning metrics of, Mean Reciprocal Rank (MRR) and Hits@1/3/10. The reported results represent the mean performance across three independent training runs, with all evaluation metrics expressed as percentages. Prior work have proposed several filtering strategies for TKG reasoning models [3]. In this study, we employ time aware filtering [4], a widely used approach for TKG extrapolation, which excludes quadruplets that are true at the query time. For studies using different filtering procedures, we report results from subsequent evaluations employing time aware filtering and exclude methods that cannot be assessed under these settings.

A.4 Transition Graph Baselines

We compare the proposed method against three settings, each evaluated using the same base architecture and training procedure. The *empirical graph* constructs successor and inhibitor edges solely from co-occurrence statistics in the training set, without incorporating any LLM-derived information. The *random graph* preserves the same number of edges and top- k structure as the LLM-derived graph, but with randomly assigned target relations and weights. Finally, the *shuffled graph* retains the topology and weight distribution of the LLM-derived graph but globally permutes the relation identities, while completely disrupting correct relation-to-relation semantics.

A.5 Pearson Correlation Analysis

To determine whether the LLM’s relational judgments simply reflect empirical co-occurrence patterns, we calculate the Pearson correlation between the transition weights assigned by the LLM and the corresponding empirical co-occurrence counts, separately for successor and inhibitor edges. Table 2 summarises the Pearson correlation values for all benchmarks.

Table 2: Pearson Correlation

Dataset	Successor edges	Inhibitor edges
ICEWS14	0.074	-0.049
ICEWS18	0.138	0.045
GDELT	0.105	0.095

Across all three datasets and both edge types, the observed correlations remain weak ($|r| \leq 0.14$), suggesting that the LLM’s successor and inhibitor judgments are largely independent of raw co-occurrence statistics rather than simply reflecting or reproducing them.

A.6 Variance across runs

Table 3 reports results over 3 independent training runs. Table 3 reports the mean, standard deviation, and 95% CI. The standard deviation is small relative to the mean across all metrics in both datasets, suggesting the results are reasonably consistent across runs.

Table 3: Variance across 3 runs. Values are mean $\pm$ std, with 95% CI in brackets (percentage points).

Dataset	MRR	H@1	H@3	H@10
ICEWS14	49.45 $\pm$ 0.15 [49.08, 49.83]	38.15 $\pm$ 0.20 [37.64, 38.66]	55.49 $\pm$ 0.15 [55.11, 55.87]	71.28 $\pm$ 0.32 [70.48, 72.07]
ICEWS18	36.02 $\pm$ 0.10 [35.78, 36.26]	24.72 $\pm$ 0.15 [24.35, 25.09]	40.74 $\pm$ 0.01 [40.71, 40.77]	58.56 $\pm$ 0.12 [58.26, 58.86]

References

1. Boschee, E., Lautenschlager, J., O’Brien, S., Shellman, S., Starz, J., Ward, M.: Icews coded event data (2015). <https://doi.org/10.7910/DVN/28075>, <https://doi.org/10.7910/DVN/28075>
2. Chen, W., Wan, H., Wu, Y., Zhao, S., Cheng, J., Li, Y., Lin, Y.: Local-global history-aware contrastive learning for temporal knowledge graph reasoning. In: 2024 IEEE 40th International Conference on Data Engineering (ICDE). pp. 733–746. IEEE (2024)
3. Gastinger, J., Szytler, T., Sharma, L., Schuelke, A., Stuckenschmidt, H.: Comparing apples and oranges? on the evaluation of methods for temporal knowledge graph forecasting. In: Joint European conference on machine learning and knowledge discovery in databases. pp. 533–549. Springer (2023)
4. Han, Z., Chen, P., Ma, Y., Tresp, V.: Explainable subgraph reasoning for forecasting on temporal knowledge graphs. In: International conference on learning representations (2020)
5. Leetaru, K., Schrod, P.A.: Gdelt: Global data on events, location, and tone, 1979–2012. In: ISA annual convention. vol. 2, pp. 1–49. Citeseer (2013)
6. Li, Z., Jin, X., Li, W., Guan, S., Guo, J., Shen, H., Wang, Y., Cheng, X.: Temporal knowledge graph reasoning based on evolutionary representation learning. In: Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval. pp. 408–417 (2021)